INTERNATIONAL CYBERSECURITY & DIGITAL FORENSICS ACADEMY

TEAM-05

Tracing a TCP/IP Communication:
Opening a Secure Website (HTTPS)

A methodical layer-by-layer architectural analysis: *"How does a single click become a secure web page?"*

STUDENT TEAM MEMBERS & REGISTRATION

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Troubleshooting Failure: Methodical TLS Diagnosis

OBSERVED ERROR

NET::ERR_CERT_DATE_INVALID

An employee accesses https://portal.company.com and the browser blocks entry: *"Your connection is not private"*.

The underlying LAN connection is fully operational — standard web traffic and email continue to function normally.

Diagnostic Goal:

Isolate the exact failing layer methodically using standard networking tools rather than guessing.

1. Physical Link	✓ip link / ifconfig UP
2. IP Connectivity	✓ping default gateway OK
3. DNS Resolution	✓nslookup resolves valid IP
4. Port Reachability	✓telnet / nc port 443 OPEN
5. TLS Handshake Probe	✗openssl s_client reports EXPIRED

Remediation:

Re-issue TLS certificate through trusted CA, bind full certificate chain on the web server daemon, reload the service, and verify via OpenSSL.

Technical References & Team Contributions

Authoritative References

IETF RFC 8446
The Transport Layer Security (TLS) Protocol Version 1.3

IETF RFC 793
Transmission Control Protocol (TCP) Specification

Cisco Networking Academy
CCNA Routing & Switching: Introduction to Networks

ICDFA Network Security Fundamentals
Module 03: Network Devices, Media, Ethernet and ARP

Layer-by-Layer Troubleshooting Method

"Can't reach https://portal.company.com — browser shows a security warning"

